PARALLEL INTERFACE

To read ADC data, or to read and write the register content over the parallel interface, tie the $\overline{\text{PAR}}$/SER SEL pin low.

The rising edge of the $\overline{\text{CS}}$ input signal three-states the bus, and the falling edge of the $\overline{\text{CS}}$ input signal takes the bus out of the high impedance state. $\overline{\text{CS}}$ is the control signal that enables the data lines and it is the function that allows multiple AD7606C-18 devices to share the same parallel data bus.

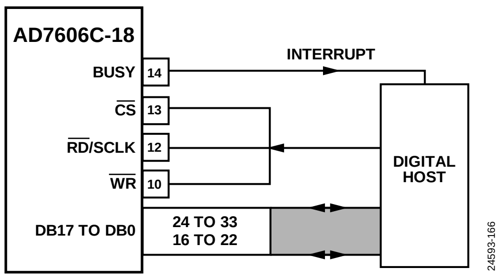

Figure 97. AD7606C-18 Interface Diagram—One AD7606C-18 Using the Parallel Bus with $\overline{\text{CS}}$ and $\overline{\text{RD}}$ Shorted Together

Reading Conversion Results (Parallel ADC Mode)

The falling edge of the $\overline{\text{RD}}$ pin reads data from the output conversion results register. Applying a sequence of $\overline{\text{RD}}$ pulses to the $\overline{\text{RD}}$ pin clocks the conversion results out from each channel onto the parallel bus, DB17 to DB0, in ascending order, from V1 to V8, as shown in Figure 98.

The parallel interface consists of 16 parallel lines on Pin 16 to Pin 22 and Pin 24 to Pin 33. Because the ADC data is 18 bit, two parallel frames are required as follows:

- 1st frame clocks out ADC data from Bit 2 to Bit 17 (MSB)
- 2nd frame clocks out ADC data from Bit 1 and Bit 0 (LSB)

The $\overline{\text{CS}}$ signal can be permanently tied low, and the $\overline{\text{RD}}$ signal can access the conversion results, as shown in Figure 3. A read operation of new data can take place after the BUSY signal goes low (see Figure 2). Alternatively, a read operation of data from the previous conversion process can take place while the BUSY pin is high.

When there is only one AD7606C-18 in a system and it does not share the parallel bus, data can be read using one control signal from the digital host. The $\overline{\text{CS}}$ and $\overline{\text{RD}}$ signals can be tied together, as shown in Figure 4. In this case, the falling edge of the $\overline{\text{CS}}$ and $\overline{\text{RD}}$ signals brings the data bus out of three-state and clocks out the data.

The FRSTDATA output signal indicates when the first channel, V1, is being read back, as shown in Figure 4. When the $\overline{\text{CS}}$ input is high, the FRSTDATA output pin is in three-state. The falling edge of $\overline{\text{CS}}$ takes the FRSTDATA pin out of three-state. The falling edge of the $\overline{\text{RD}}$ signal corresponding to the result of V1 sets the FRSTDATA pin high, indicating that the result from V1 is available on the output data bus. The FRSTDATA pin returns to a logic low following the next falling edge of $\overline{\text{RD}}$.

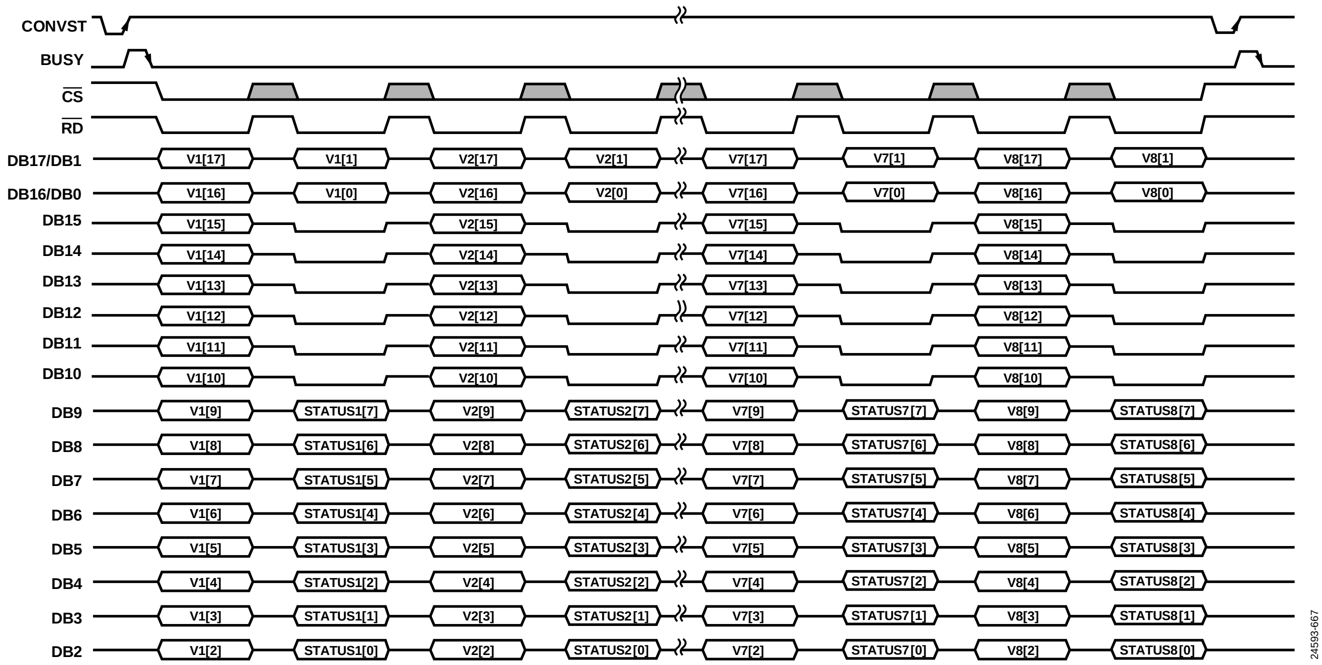

Figure 98. Parallel Interface, ADC Mode with Status Header Enabled

Reading During Conversion

Data read operations from the AD7606C-18, as shown in Figure 99, can occur in the following three scenarios:

- After a conversion while the BUSY line is low
- During a conversion while the BUSY line is high
- Starting while the BUSY line is low and ending while the following conversion is in progress, see Figure 2

Reading during conversions has little effect on the performance of the converter, and it allows a faster throughput rate to be achieved. Data can be read from the AD7606C-18 at any time other than on the falling edge of the BUSY signal because this is when the output data registers are updated with the new conversion data. Any data read while the BUSY signal is high must be completed before the falling edge of the BUSY signal.

Parallel ADC Mode with CRC Enabled

In software mode, the parallel interface supports reading the ADC data with the CRC appended, when enabled through the INT_CRC_ERR_EN bit (Address 0x21, Bit 2). The CRC is 16 bits, and it is clocked out after reading all eight channel conversions, as shown in Figure 100. The CRC calculation includes all data on the DBx pins: data, status (when appended), and zeros. See the Diagnostics section for more details on CRC.

Parallel ADC Mode with Status Enabled

In software mode, the 8-bit status header is enabled (see Table 25) by setting Bit 6 in the CONFIG register (Address 0x02, Bit 6), and each channel then takes the following two frames of data:

- The first frame clocks the ADC data out normally through DB17 to DB2 from the MSB to Bit 2.
- The second frame clocks out the status header of the channel on DB9 to DB2, DB9 being the MSB and DB2 being the LSB of the status header, while DB1 to DB0 clock out the two LSBs of the conversion result and the DB15 to DB10 pins clock out zeros.

This sequence is shown in Figure 98. Table 25 explains the status header content and describes each bit.

Table 24. CH.ID Bits Decoding in Status Header

CH.ID2	CH.ID1	CH.ID0	Channel Number
0	0	0	Channel 1 (V1)
0	0	1	Channel 2 (V2)
0	1	0	Channel 3 (V3)
0	1	1	Channel 4 (V4)
1	0	0	Channel 5 (V5)
1	0	1	Channel 6 (V6)
1	1	0	Channel 7 (V7)
1	1	1	Channel 8 (V8)

Table 25. Status Header, Parallel Interface

	Bit 7 (MSB)	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0 (LSB)
Content	RESET_DETECT	DIGITAL_ERROR	OPEN_DETECTED	RESERVED		CH. ID 2	CH. ID 1	CH. ID 0
Meaning[1]	Reset detected	Error flag on Address 0x22	The analog input of this channel is open			Channel ID (see Table 24)		

[1] See the Diagnostics section for more information.

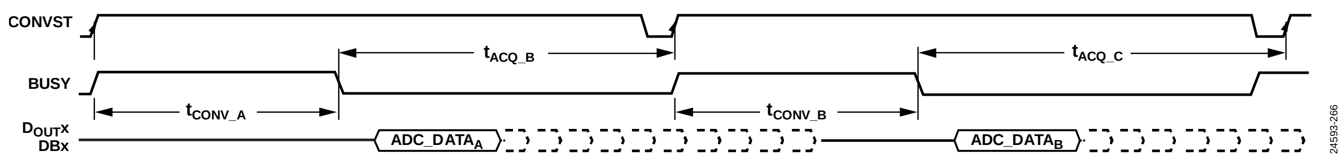

Figure 99. ADC Data Read Can Happen After Conversion and/or During the Following Conversion

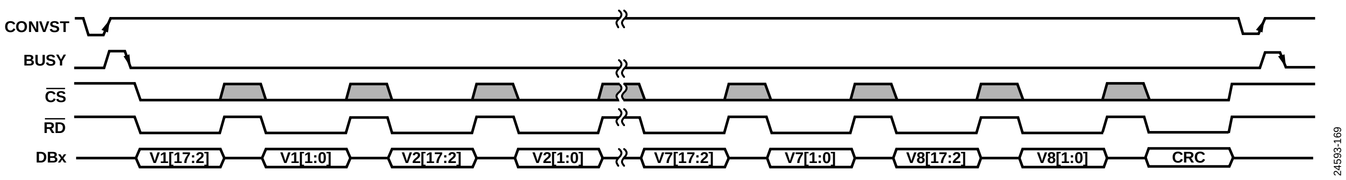

Figure 100. Parallel Interface, ADC Mode with CRC Enabled

Parallel Register Mode (Reading Register Data)

In software mode, all of the registers in Table 31 can be read over the parallel interface. Bits[DB17:DB2] leave a high impedance state when both the $\overline{\text{CS}}$ signal and $\overline{\text{RD}}$ signal are logic low for reading register content, or when both the $\overline{\text{CS}}$ signal and $\overline{\text{WR}}$ signal are logic low for writing register address and/or register content.

A register read is performed through two frames: first, a read command is sent to the AD7606C-18 and second, the AD7606C-18 clocks out the register content. The format for a register read command is shown in Figure 101. On the first frame, perform the following:

- Bit DB17 must be set to 1 to select a read command. The read command puts the AD7606C-18 into register mode.
- Bits[DB16:DB10] must contain the register address.
- The subsequent eight bits, Bits[DB9:DB2], are ignored.

The register address is latched on the AD7606C-18 on the rising edge of the $\overline{\text{WR}}$ signal. The register content can then be read from the latched register by bringing the $\overline{\text{RD}}$ line low on the following frame, as follows:

- Bit DB17 is pulled to 0 by the AD7606C-18.
- Bits[DB16: DB10] provide the register address being read.
- The subsequent eight bits, Bits[DB9: DB2], provide the register content.

To revert to ADC mode, keep all DBx pins low during one $\overline{\text{WR}}$ cycle, as shown in the Parallel Register Mode (Writing Register Data) section. No ADC data can be read while the device is in register mode.

Parallel Register Mode (Writing Register Data)

In software mode, all of the R/W registers in Table 31 can be written to over the parallel interface. To write a sequence of registers, exit ADC mode (default mode) by reading any register on the memory map. A register write command is performed by a single frame, via the parallel bus (Bits[DB17:DB2]), $\overline{\text{CS}}$ signal, and $\overline{\text{WR}}$ signal. The format for a write command, as shown in Figure 101, is structured as follows:

- Bit DB17 must be set to 0 to select a write command.
- Bits[DB16:DB10] contain the register address.
- The subsequent eight bits, Bits[DB9:DB2], contain the data to be written to the selected register.

Data is latched onto the device on the rising edge of the $\overline{\text{WR}}$ pin. To revert back to ADC mode, keep all DBx pins low during one $\overline{\text{WR}}$ cycle. No ADC data can be read while the device is in register mode.

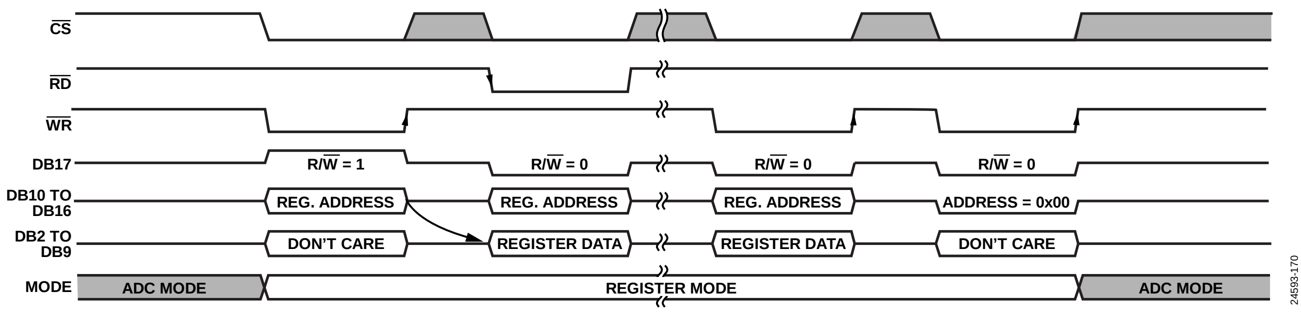

Figure 101. Parallel Interface Register Read Operation Followed by a Write Operation

REGISTER SUMMARY

Table 31. AD7606C-18 Register Summary

Addr	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Reset	R/W
0x01	STATUS	RESET_DETECT	DIGITAL_ERROR	OPEN_DETECTED	RESERVED					0x00	R
0x02	CONFIG	RESERVED	STATUS_HEADER	EXT_OS_CLOCK	DOUT_FORMAT		RESERVED	OPERATION_MODE		0x08	R/W
0x03	RANGE_CH1_CH2	CH2_RANGE				CH1_RANGE				0x33	R/W
0x04	RANGE_CH3_CH4	CH4_RANGE				CH3_RANGE				0x33	R/W
0x05	RANGE_CH5_CH6	CH6_RANGE				CH5_RANGE				0x33	R/W
0x06	RANGE_CH7_CH8	CH8_RANGE				CH7_RANGE				0x33	R/W
0x07	BANDWIDTH	CH8_BW	CH7_BW	CH6_BW	CH5_BW	CH4_BW	CH3_BW	CH2_BW	CH1_BW	0x00	R/W
0x08	OVERSAMPLING	OS_PAD				OS_RATIO				0x00	R/W
0x09	CH1_GAIN	RESERVED		CH1_GAIN						0x00	R/W
0x0A	CH2_GAIN	RESERVED		CH2_GAIN						0x00	R/W
0x0B	CH3_GAIN	RESERVED		CH3_GAIN						0x00	R/W
0x0C	CH4_GAIN	RESERVED		CH4_GAIN						0x00	R/W
0x0D	CH5_GAIN	RESERVED		CH5_GAIN						0x00	R/W
0x0E	CH6_GAIN	RESERVED		CH6_GAIN						0x00	R/W
0x0F	CH7_GAIN	RESERVED		CH7_GAIN						0x00	R/W
0x10	CH8_GAIN	RESERVED		CH8_GAIN						0x00	R/W
0x11	CH1_OFFSET	CH1_OFFSET								0x80	R/W
0x12	CH2_OFFSET	CH2_OFFSET								0x80	R/W
0x13	CH3_OFFSET	CH3_OFFSET								0x80	R/W
0x14	CH4_OFFSET	CH4_OFFSET								0x80	R/W
0x15	CH5_OFFSET	CH5_OFFSET								0x80	R/W
0x16	CH6_OFFSET	CH6_OFFSET								0x80	R/W
0x17	CH7_OFFSET	CH7_OFFSET								0x80	R/W
0x18	CH8_OFFSET	CH8_OFFSET								0x80	R/W
0x19	CH1_PHASE	CH1_PHASE								0x00	R/W
0x1A	CH2_PHASE	CH2_PHASE								0x00	R/W
0x1B	CH3_PHASE	CH3_PHASE								0x00	R/W
0x1C	CH4_PHASE	CH4_PHASE								0x00	R/W
0x1D	CH5_PHASE	CH5_PHASE								0x00	R/W
0x1E	CH6_PHASE	CH6_PHASE								0x00	R/W
0x1F	CH7_PHASE	CH7_PHASE								0x00	R/W
0x20	CH8_PHASE	CH8_PHASE								0x00	R/W
0x21	DIGITAL_DIAG_ENABLE	INTERFACE_CHECK_EN	CLK_FS_OS_COUNTER_EN	BUSY_STUCK_HIGH_ERR_EN	SPI_READ_ERR_EN	SPI_WRITE_ERR_EN	INT_CRC_ERR_EN	MM_CRC_ERR_EN	ROM_CRC_ERR_EN	0x01	R/W
0x22	DIGITAL_DIAG_ERR	RESERVED		BUSY_STUCK_HIGH_ERR	SPI_READ_ERR	SPI_WRITE_ERR	INT_CRC_ERR	MM_CRC_ERR	ROM_CRC_ERR	0x00	R/W
0x23	OPEN_DETECT_ENABLE	CH8_OPEN_DETECT_EN	CH7_OPEN_DETECT_EN	CH6_OPEN_DETECT_EN	CH5_OPEN_DETECT_EN	CH4_OPEN_DETECT_EN	CH3_OPEN_DETECT_EN	CH2_OPEN_DETECT_EN	CH1_OPEN_DETECT_EN	0x00	R/W
0x24	OPEN_DETECTED	CH8_OPEN	CH7_OPEN	CH6_OPEN	CH5_OPEN	CH4_OPEN	CH3_OPEN	CH2_OPEN	CH1_OPEN	0x00	R/W
0x28	DIAGNOSTIC_MUX_CH1_2	RESERVED		CH2_DIAG_MUX_CTRL			CH1_DIAG_MUX_CTRL			0x00	R/W
0x29	DIAGNOSTIC_MUX_CH3_4	RESERVED		CH4_DIAG_MUX_CTRL			CH3_DIAG_MUX_CTRL			0x00	R/W
0x2A	DIAGNOSTIC_MUX_CH5_6	RESERVED		CH6_DIAG_MUX_CTRL			CH5_DIAG_MUX_CTRL			0x00	R/W
0x2B	DIAGNOSTIC_MUX_CH7_8	RESERVED		CH8_DIAG_MUX_CTRL			CH7_DIAG_MUX_CTRL			0x00	R/W
0x2C	OPEN_DETECT_QUEUE	OPEN_DETECT_QUEUE								0x00	R/W
0x2D	FS_CLK_COUNTER	CLK_FS_COUNTER								0x00	R
0x2E	OS_CLK_COUNTER	CLK_OS_COUNTER								0x00	R
0x2F	ID	DEVICE_ID				SILICON_REVISION				0x31	R